

# Senior Backend Engineer — Take-Home Exercise

## Scenario

You are a senior backend engineer at Amazon. A product manager needs a proof of concept built quickly: it will go live and handle a small fraction of real customer traffic. Due to project constraints, you are the only engineer on it.

The PoC is very likely to be taken further, and the team is expected to grow around it. Build accordingly — pragmatic enough to ship now, sound enough that other engineers can pick it up and extend it.

**Deliverable:** a GitHub repository. Include a README explaining your decisions, trade-offs, and anything you'd do differently with more time.

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## Task 1 — File-zipping REST API

Build a REST API in Python using FastAPI with a single endpoint that:

- accepts several uploaded files in one request
- zips them
- returns the zip archive as the response

Consider what a live endpoint needs: validation, error handling, sensible behaviour on large or numerous files, and appropriate response headers. Tests are expected.

## Task 2 — Containerisation

Provide a Docker image for the service. It should be straightforward for another engineer to build and run locally from the repository.

## Task 3 — CI/CD pipeline

Set up a CI/CD pipeline (GitHub Actions or a tool of your choice, with the config committed to the repo). Define what should run automatically and why, given a codebase that will soon have multiple contributors.

## Task 4 — AI-assisted development (written answer)

In a markdown file in the repository, describe your ideal way of developing software with AI assistance. Cover:

- which AI tools you use and where each fits in your workflow
- how you guarantee quality throughout the process
- how this changes when the team grows from one engineer to several

Answer from your own practice — we're interested in how you actually work, not a survey of the tooling landscape.

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## **What we're assessing**

Code quality and structure, judgement about what matters for a PoC that will grow, operational thinking (containerisation, automation, failure modes), clarity of written communication, and a considered rather than reflexive approach to AI tooling.